2022 Ph H2 Q3

Section: Our Dynamic Universe

Topic: Collisions, Explosions and Impulse

Two trolleys move together at 0.40 m s^{-1}. Mass of X = 0.50 kg, Y = 0.25 kg. A plunger in X pushes Y with average force 6.25 N. After the push, Y moves at 1.80 m s^{-1}. (a) (i) Calculate the change in momentum of Y, (ii) the contact time. (b) Find the velocity of X immediately after. (c) Explain how to test whether the interaction is elastic. (d) Photodiode questions: (i) name the effect, (ii) explain using band theory.

Worked solution

(a)(i) Change in momentum of Y

$\Delta p = m(v - u) = 0.25 \times (1.80 - 0.40)$
$= 0.350$ kg m s^{-1}

Answer: 0.350 kg m s^{-1}

(a)(ii) Contact time

Impulse $J = F\,\Delta t = \Delta p$
$\Delta t = \Delta p / F = 0.350 / 6.25 = 0.056$ s

Answer: 0.056 s

(b) Velocity of X after separation

Conservation of momentum:

Initial p = (0.50 + 0.25)×0.40 = 0.300 kg m s^{-1}
Final p = (0.50)v_X + (0.25×1.80)
0.300 = 0.50 v_X + 0.450
v_X = (0.300 − 0.450)/0.50 = -0.30 m s^{-1}

Answer: v_X = −0.30 m s^{-1} (to the left)

(c) Testing for elastic collision

Measure the kinetic energy before and after the interaction.
• If total KE is conserved, the collision is elastic.
• If KE is reduced, it is inelastic.

(d)(i) The effect is the photovoltaic effect.

(d)(ii) Band theory explanation

• In the photodiode (a p–n junction), photons incident on the material have energy ≥ band gap.
• Electrons are excited from the valence band to the conduction band, creating electron–hole pairs.
• The junction electric field separates these charges, producing a potential difference.

Final answers

(a)(i) Δp_Y = 0.350 kg m s^{-1}

(a)(ii) Δt = 0.056 s

(b) v_X = −0.30 m s^{-1} (to the left)

(c) Check KE before and after — conserved = elastic

(d)(i) Photovoltaic effect

(d)(ii) Photon excites electron across band gap; junction field separates charges.

Revision tips

- Impulse = change in momentum = $F\Delta t$.
- Conservation of momentum applies when no external forces act.
- Elastic collision ⇔ total kinetic energy conserved.
- Photodiode works via photovoltaic effect — photons create e^-/h^+ pairs.
- Band theory: photon energy promotes electron across band gap.